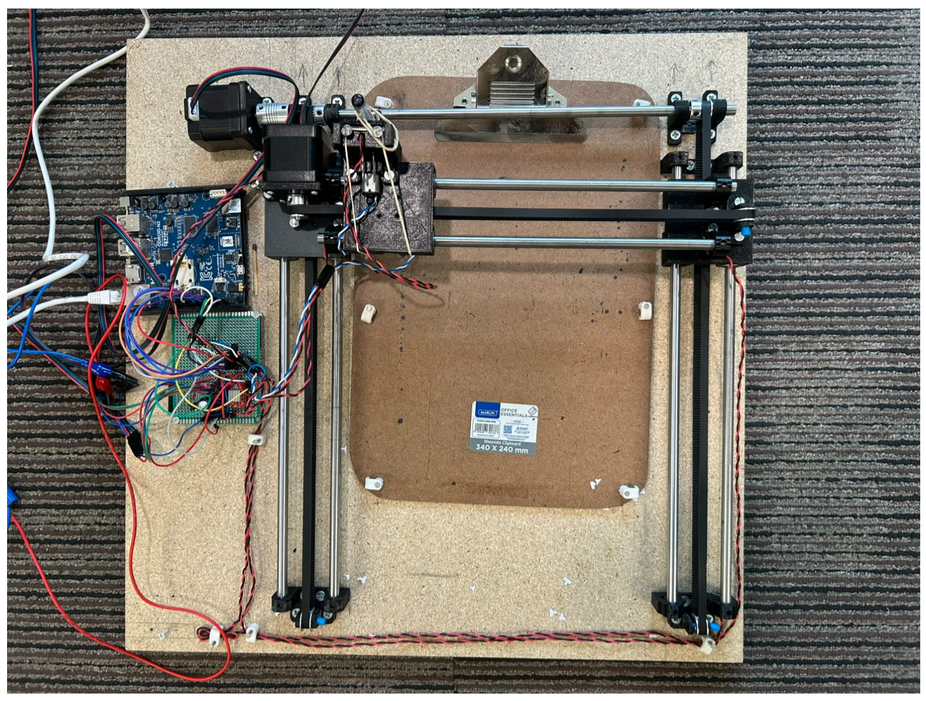
Handwriting robot

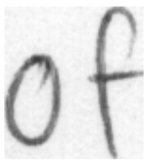
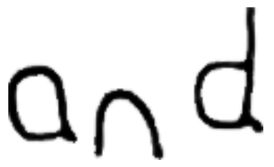
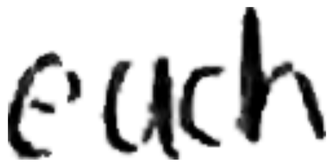
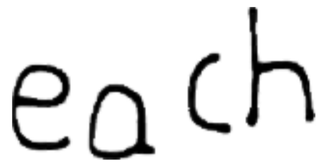
Recognise a page. Reproduce an individual writing style.

D.R. Braune | 23539195 | Supervisor: Prof. H.C. Myburgh

THE BUILT SYSTEM



Cartesian belt drive, separate pen lift and ODROID control.

	ORIGINAL SAMPLE	ROBOT WRITING
150		
O		
Y		

Actual word photographs show the style retained in physical output.

Full system functional flow

PREPARE	**Labelled handwriting** →	**Recognition models** →	**Stored glyph profiles**
READ	**Camera photograph** →	**Condition and split lines** →	**Text + author**
REPRODUCE	**New text + profile** →	**Glyph paths + G-code** →	**Gantry + written page**

Models and profiles are prepared offline, then reused during operation.

Subsystem design and visual checks

01 Page conditioning and segmentation

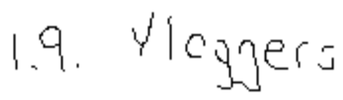
Correct lighting and rotation.
Assign ink to ordered line crops.



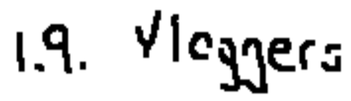
02 Text and author recognition

Read complete lines with CNN–BiLSTM–CTC.
Normalise stroke width for author identification.

Centreline



Fixed width, 0.17 x-height



03 Glyph extraction

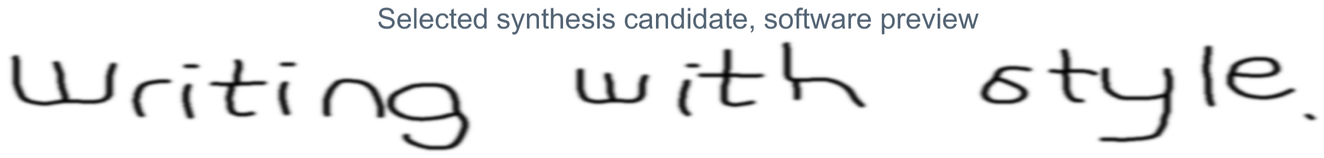
Align known text, refine character cuts
and trace stored centreline paths.



04 Synthesis and G-code

Select variants, place and join glyphs,
then map paths into machine coordinates.

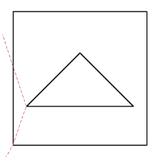
Selected synthesis candidate, software preview



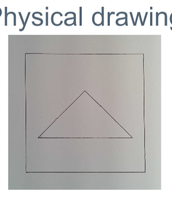
05 Motion and pen control

Coordinate two stepper axes and the pen lift.
Use switches for homing and interruption.

Commanded path

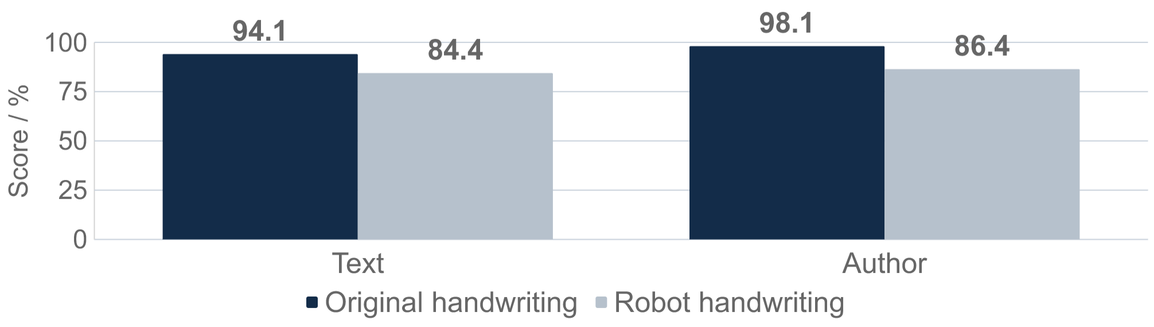


Physical drawing

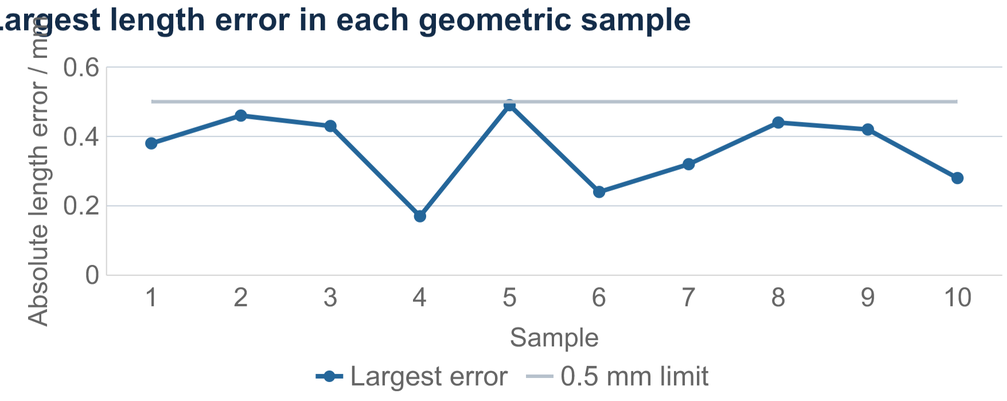


Measured performance

Reader and author classifier scores



Largest length error in each geometric sample



11 styles

70 / 70 lengths ≤ 0.5 mm

25 / 25 jobs ≤ 4 s / character

Main limits are connected-letter context and classification time.

Faculty of Engineering, Built Environment and Information Technology
Fakulteit Ingenieurswese, Bou-omgewing en Inligtingtegnologie / Lefapha la Boetšenere, Tikologo ya Kago le Theknolotši ya Tshedimošo
UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Department of Electrical, Electronic and Computer Engineering

Methods: Graves et al. (2006), Haines et al. (2016), Zhang and Suen (1984).
Project records: original validation, 14 physical pages and 10 geometric samples.
Geometric results describe feature lengths. Classification latency remains above target.